LESSON 7.5

Represent Rational Numbers on the Coordinate Plane

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can plot and identify points on the coordinate plane using ordered pairs.

WHY THIS MATTERS Captain Vega left a treasure map, but instead of an 'X marks the spot,' she hid the directions as ordered pairs! To find the treasure, you need to plot each point on the coordinate plane and connect them in order. The first clue says: Start at (2, 1), then go to (5, 4), then (8, 1).

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Coordinate plane *Spanish: Plano cartesiano*	A grid with a line going across and a line going up to plot points.	x-axis across, y-axis up
Ordered pair *Spanish: Par ordenado*	Two numbers (x, y) that tell where a point is on a grid.	(−3, 4) is an ordered pair
Origin *Spanish: Origen*	The point (0, 0) where the two grid lines cross.	every plot starts here
Quadrant *Spanish: Cuadrante*	One of the four parts of a coordinate grid, numbered I to IV.	Quadrant II is upper left
Reflection *Spanish: Reflexión*	A flipped, mirror image across a line.	(3, 2) reflected over the x-axis becomes (3, -2) — same x, opposite y
Integer *Spanish: Número entero*	Whole numbers and their opposites, like -2, -1, 0, 1, 2.	..., -3, -2, -1, 0, 1, 2, 3, ...

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Plot (−5, −4). Name its quadrant.

1. Start at the origin, (0, 0).
2. Move 5 units left, because x = −5.
3. Move 4 units down, because y = −4.

Answer: (−, −) is Quadrant III

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's plot (2, 1). Where do we start? At the origin (0, 0).

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The ordered pair _____ tells me to move _____ and then _____ .
- The point _____ lands in Quadrant _____ because its signs are _____ .
- I always start at the origin because _____ .
- A point sitting on the _____ -axis has a coordinate of _____ .

Word bank coordinate plane | ordered pair | origin | quadrant | x-axis | y-axis | x-coordinate | y-coordinate | horizontal | vertical | left | right

WATCH OUT!

A common mistake

A common mistake in Graph on the Coordinate Plane is swapping the order of the coordinates

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

A point is at (6, 2). What does the 6 tell you?

(A) Move 6 units to the right
(B) Move 6 units up
(C) Move 6 units to the left
(D) The point is in Quadrant 6

STEPS TO TRY

1. In (6, 2), the 6 is the FIRST number. First and second positions have different jobs.
2. First number = x = horizontal (right/left). Second number = y = vertical (up/down). Positive means right or up.
3. The 6 sits in the x spot and it's positive, so it describes a horizontal move. Which direction is positive x?

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Put these steps in order for plotting the point (4, 5).

Groups:

_____ Start at the origin (0, 0)

_____ Move 4 units to the right along the x-axis

_____ Move 5 units up from that spot

_____ Place a dot at the point (4, 5)

STEPS TO TRY

1. Every plotting routine begins at the same home spot before any moving happens.
2. Think: start point, then horizontal move, then vertical move, then mark the dot. Order the cards to tell that story.
3. Place 'Start at the origin (0, 0)' first. Then ask: does the 4 (x) move or the 5 (y) move come next?

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Which ordered pair names a point 3 units right and 7 units up from the origin?

(A) (0, 7)

(B) (3, 7)

(C) (3, 3)

(D) (7, 3)

STEPS TO TRY

1. Re-read the moves: 3 units right, then 7 units up. Which move comes first in an ordered pair?
2. An ordered pair is (x, y): x is the right/left distance, y is the up/down distance. Match each move to its slot.
3. Start at the origin and go right 3 — that 3 is the x, the first number. Now decide where the 'up 7' goes.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What are the coordinates of the origin?

(A) (0, 0)

(B) (0, 1)

(C) (1, 1)

(D) (1, 0)

STEPS TO TRY

1. The origin is the exact spot where the x-axis and y-axis cross. Picture that crossing point.
2. Ask: how far right have you moved at the origin? How far up? Those two distances are the coordinates.
3. At the crossing point you haven't moved right at all, so x = 0. Now figure out the y-value the same way.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

You plotted a triangle. How did the ordered pairs tell you exactly where to place each point?

SENTENCE STARTER The first number in the ordered pair tells me ___ and the second number tells me ___. The point (5, 4) means ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help